By least square method, we can find the matrix T and translation vector b correspondingly. As a example, for $Goldfish$ label, the minimum error and other related data for two cases can be found in TABLE II.

Mean value of functions	$\mathfrak{C}_6$	$\mathfrak{C}_7$	$\mathfrak{C}_8$
$\|\|\mathbf{v}_i^1\|\|_2$	270	94	103
$\|\|\mathbf{v}_i^2\|\|_2$	249	94	103
$\|d_{ij}^1 - d_{ij}^2\|$	0.0039	0.0001	0.0003
$d(\mathbf{v}_i^1, \mathbf{v}_i^2)$	350	122	48
$\|\|\mathcal{N}(\mathbf{v}_i^1)\|\|_2$	27	29	31
$\|\|\mathcal{N}(\mathbf{v}_i^2)\|\|_2$	24	29	31
$d(\mathcal{N}(\mathbf{v}_i^1), \mathcal{N}(\mathbf{v}_i^2))$	36	42	42
$r(\mathcal{N}(X_J^1))$	738	865	1000
$r(\mathfrak{T}(\mathcal{N}(X_J^1)))$	618	860	999
L_{O-RMSE}	0.76	0.98	0.92
$\|\|\mathcal{N}(\mathcal{RD}(\mathbf{v}_i^1))\|\|_2$	3.70	3.56	3.11
$\|\|\mathcal{N}(\mathcal{RD}(\mathbf{v}_i^2))\|\|_2$	3.70	3.56	3.11
$d(\mathcal{N}(\mathcal{RD}(\mathbf{v}_i^1)), \mathcal{N}(\mathcal{RD}(\mathbf{v}_i^2)))$	5.17	5.00	4.35
$r(\mathcal{N}(\mathcal{RD}(X_J^1)))$	14	13	10
$r(\mathfrak{T}(\mathcal{N}(\mathcal{RD}(X_J^1))))$	14	13	10
$L_{RD-RMSE}$	1.77	1.96	1.93

TABLE II: The length $\|\cdot\|_2$, rank r, distance d_{ij} and RMSE L for each fully connected layer of normalized data for two cases, where $\mathfrak{C} = goldfish$.

In the first case, the results show that (for every layer and all considered patches), L_{O-RMSE} is small comparing to $d(\mathcal{N}(\mathbf{v}_i^1), \mathcal{N}(\mathbf{v}_i^2))$, yet the rank of $\mathfrak{T}(\mathcal{N}(X_J^1))$ will decrease a little compared to that of $\mathcal{N}(X_J^1)$. In the second case, though the rank of $\mathfrak{T}(\mathcal{N}(\mathcal{RD}(X_J^1)))$ equals the rank of $\mathcal{N}(\mathcal{RD}(X_J^1))$, $L_{RD-RMSE}$ between $\mathcal{N}(\mathcal{RD}(X_J^1))$ and $\mathcal{N}(\mathcal{RD}(X_J^2))$ is nearly 40% of the distance of $d(\mathcal{N}(\mathcal{RD}(\mathbf{v}_i^1)), \mathcal{N}(\mathcal{RD}(\mathbf{v}_i^2)))$. Thus, this indicates that for each point $\mathbf{v}_i$, the projection of its patch to the tangent space at $\mathbf{v}_i$ behaves differently separately for two Alexnets, but the patches on the manifold for two Alexnets are somehow similar.

From the experiment and analysis in this section, we can conclude that as far as manifolds are concerned, two AlexNets do behave consistently in some aspects and sharing some information, for example, similar activation vectors' length, similar distance matrix, small RMSE, similar Euclidean distance and same dimensions. On the other hand, the distinct behaviors of the projection of patches to the tangent space have revealed the existence of some differences. Yet a simple linear transformation can neither identify nor explain the difference. Moreover, the manifold itself could be highly curved. In order to compare two highly curved manifolds in high dimension and identify the similarities and differences, we take the intrinsic geometry of a manifold into consideration.

B. Computation of Riemann curvature

Based on the results in Section II-B and II-D, we can compare the Riemann curvature tensor of $\mathcal{M}$ with that of ambient space $\widetilde{\mathcal{M}}$. Here we make use of the Gauss equation which shows the relationship between the Riemann curvature tensor of sub-manifold and that of ambient space.

Theorem 1 (The Gauss Equation [12]): Let $\mathcal{M}$ be an d-dimensional embedded submanifold of a Riemannian manifold $\widetilde{\mathcal{M}}$ of dimension D, for any vector fields $X, Y, Z, W \in T\mathcal{M}$, the following equation holds:

$$\widetilde{\mathcal{R}}(X,Y,Z,W) = \mathcal{R}(X,Y,Z,W) - \langle \beta(X,W), \beta(Y,Z) \rangle + \langle \beta(X,Z), \beta(Y,W) \rangle.$$

where $\beta(X,Y) = \widetilde{\nabla}_X Y - \nabla_X Y$ is the second fundamental form, $\widetilde{\mathcal{R}}$ is the Riemann curvature tensor of $\widetilde{\mathcal{M}}$ and $\mathcal{R}$ is that of $\mathcal{M}$.

Similar to the work in [11], in our case, the Riemannian manifold $\widetilde{\mathcal{M}}$ is Euclidean space $\mathbb{R}^D$, so $\widetilde{\mathcal{R}}(X,Y,Z,W) = 0$, then, the Riemann curvature of $\mathcal{M}$ is represented as

$$\mathcal{R}(X,Y,Z,W) = \langle \beta(X,W), \beta(Y,Z) \rangle - \langle \beta(X,Z), \beta(Y,W) \rangle.$$

As for the second fundamental form, we can interpret it as a measure of the difference between the Riemannian connection on $\mathcal{M}$ and $\widetilde{\mathcal{M}}$. Because Riemann curvature tensor is linear with respect to X, Y, Z, W, we only need to determine it in terms of a basis of $T_{\mathbf{v}}\mathcal{M}$ for each point $\mathbf{v}$, then we construct a local natural orthonormal coordinate frame $\{\frac{\partial}{\partial x^1}, \cdots, \frac{\partial}{\partial x^d}, \frac{\partial}{\partial y^1}, \cdots, \frac{\partial}{\partial y^{D-d}}\}$ of the ambient space $\widetilde{\mathcal{M}}$ at point $\mathbf{v}$, the restrictions of $\{\frac{\partial}{\partial x^1}, \cdots, \frac{\partial}{\partial x^d}\}$ to $\mathcal{M}$ form a local orthonormal frame of $T_{\mathbf{v}}\mathcal{M}$, the last $D-d$ orthonormal coordinates $\{\frac{\partial}{\partial y^1}, \cdots, \frac{\partial}{\partial y^{D-d}}\}$ form a local orthonormal frame of $\mathcal{N}_{\mathbf{v}}\mathcal{M}$. Then, we aim at computing $\mathcal{R}(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^l}, \frac{\partial}{\partial x^j}, \frac{\partial}{\partial x^k})$, denoted as R_{iljk}.

Under this locally natural orthonormal coordinate frame, the embeding map f is redefined as

$$f(x^1, x^2, \cdots, x^d) = [x^1, x^2, \cdots, x^d, f^1, f^2, \cdots, f^{D-d}].$$

where $x = [x^1, x^2, \cdots, x^d]$ are natural parameters. Then the second fundamental form is shown as

$$\beta(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j}) = \sum_{\alpha=1}^{D-d} h_{ij}^{\alpha} \frac{\partial}{\partial y^{\alpha}},$$

with $h_{ij}^{\alpha}, (\alpha = 1, \cdots, D-d)$ being the second derivative $\frac{\partial^2}{\partial x^i \partial x^i}$ of embedding component function f^{α}, which constitutes the Hessian matrix $H^{\alpha} = \left(\frac{\partial^2}{\partial x^i \partial x^j}\right)$, correspondingly, the Riemann curvature of $\mathcal{M}$ is represented as:

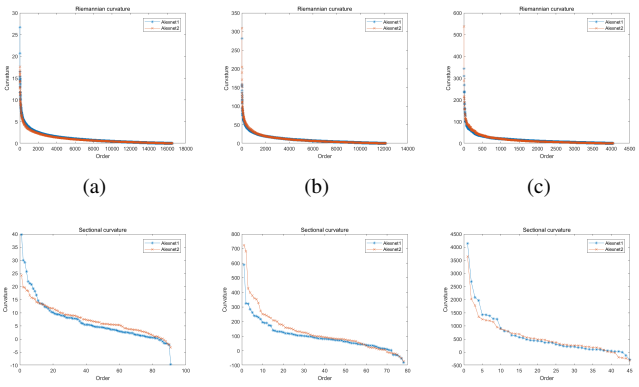
$$R_{iljk} = \sum_{\alpha=1}^{D-d} (h_{ik}^{\alpha} h_{lj}^{\alpha} - h_{ij}^{\alpha} h_{lk}^{\alpha}).$$

Note that Hessian matrix is a square matrix of second-order derivatives with respect to all of the variables of scalar-values function. It can represent the concavity, convexity and the local curvature of a function. Suppose $f^{\alpha}(x^1, x^2, \cdots, x^d)$ is a multivariate function with d parameters. Then the Hessian matrix H of f^{α} is given as $H_{ij}(f^{\alpha}) = \left(\frac{\partial^2}{\partial x^i \partial x^j}\right)$. It follows that to compute the Riemann curvature of Riemannian submanifold $\mathcal{M}$, we just need to estimate the Hessian matrix of the embedding map f.

We follow the strategy in [11] to estimate the Hessian matrix. Finally we are able to calculate the Riemann curvature tensor.

C. Experiments

Applying the method mentioned above, we can get the Riemann curvature tensor of all fully connected layers. As $R_{ijlk} = -R_{jilk} = -R_{ijkl} = R_{lkij}$, we only calculate the positive part. At the same time, in order to do the comparison, we sort the tensor as a vector in descending order (for the correspondence of axs between two manifolds is unknown). With the computation of Riemann curvature tensor, the sectional curvatures can also be easily calculated and sorted in descending order. As an example, one result of curvature distribution curves is shown in Fig. 2.



(a) (b) (c)

(d) (e) (f)

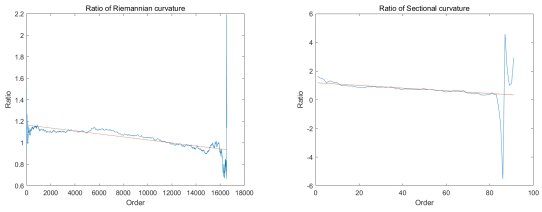
Fig. 2: For a choosen $Goldfish$ image, (a), (b), (c) are the Riemann curvature distributions of manifolds $\mathfrak{C}_6$,$\mathfrak{C}_7$ and $\mathfrak{C}_8$ respectively. (d), (e), (f) are the sectional curvature distributions of manifolds $\mathfrak{C}_6$,$\mathfrak{C}_7$ and $\mathfrak{C}_8$ respectively. It is obvious that the curvature distributions for corresponding fully connected layer almost coincide with each other for two AlexNets.

D. Analysis

In order to build a statistical result, we need to introduce a quantified characterization of the similarities of two Riemann (sectional resp.) curvature distribution curves. Here we use similar ratio. Actually, as the magnitude of the curvature ranges from 0 to 10^3, we cannot just use the distance between two vectors to justify the similarity. Instead, for two curvature distribution vectors $\{a_1, a_2, \cdots, a_n\}$ and $\{b_1, b_2, \cdots, b_n\}$, we calculate the ratio vector $\{r_1, r_2, \cdots, r_n\}$ where $r_i = \frac{a_i}{b_i}$. Least square method is applied to find the optimum fitting straight line $r = ki + b$ and we define $r_0 = \frac{1}{2}kn + b$ as the similar ratio of these two curvature distribution vectors. The closer r_0 is to 1, the more similar two curvature distributions are. An example of the ratio curve of Riemann curvature distribution, sectional curvature distribution and the optimum fitting straight line is given in Fig.3.

As similar ratio reflects the similarity between images, in TABLE III we give the percentage of total images whose similar ratio lies in the specified range.

The percent of similar ratio in $[0.8, 1.2]$ shows that the Riemann curvature distributions and sectional curvature distributions in two Alexnets are nearly the same for a majority of images. This indicates that manifold are shaped in the same way at these points. Meanwhile, we can find that there is a correspondence between axs of the tangent space of two manifolds at corresponding points. We thus conclude that for these images, the two Alexnets are extracting the same features of the images to classify them. On the other hand, for some other images, there are pieces of Riemann curvature and sectional curvature that don't match well such as the example in Fig. 4. This indicates that for these images, two Alexnets are extracting different features though finally they still have the same classfication. To identify these difference is part of our future work.

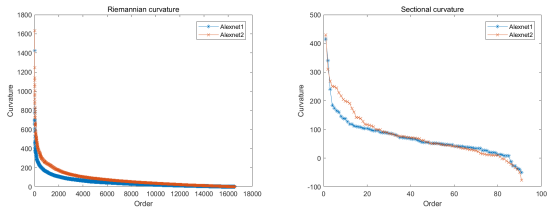


(a) (b)

Fig. 3: (a),(b) are Ratio curve of curves Fig.2 (a),(d). The optimum fitting straight line is painted in red. It can be seen that the similar ratio do show the similarities of two distribution.

Range	$\mathfrak{C}_6$	$\mathfrak{C}_7$	$\mathfrak{C}_8$
$0.8 < r_{Rie} < 1.2$	66%	56%	42%
$0.8 < r_{Sec} < 1.2$	30%	26%	14%

TABLE III: The percent of similar ratio in range $[0.8, 1.2]$ of every fully connected layer, where r_{Rie} (r_{Sec}) stands for the similar ratio for Riemann (sectional) curvature distribution and $\mathfrak{C} = Goldfish$.



(a) (b)

Fig. 4: (a), (b) are Riemann curvature distribution and sectional curvature distribution of fully connected layers where difference is relative large.

IV. Related Work

The conception of curvature-aware methods have been implemented in dimension reduction. Traditional manifold learning algorithms, including LLE, LEP, LTSA, PFE [10] et. al., always assume that the manifold is locally isometric to the Euclidean space. However, as being highlighted in [11], manifold $\mathcal{M}$ could be highly curved and not isometric to Euclidean space. In [11], the authors have proposed a curvature-aware manifold learning algorithm and achieve better results.

Comparing to our work, we include curvature tensor as a criteria for comparing neural networks.

Next, we are building a connection between manifolds and artificial neural networks. In [13], Braham et al. consider the manifold structure of the data across the layers of a deep learning network. They have defined a set of measures which has the intrinsic assumption that the manifolds are isometric to Euclidean space. Basri et al. [14] consider the ability and efficiency of deep neural networks to represent data that lies near a low-dimensional manifold in a high-dimensional space. Other researchers like Rifai et al. [15], [16] mainly focus on representing the atlas of manifold charts. Comparatively, we use the well-known AlexNet and put no special assumption on the manifolds. Our dimension decreasing results in Section II-D could be seen as an evidence to Basri et al.'s work.

Then, for the comparison of two given neural networks, most of the known work focus on comparing the architecture and performance, for example [17], [18], [19]. The work in [4], [20] focus on comparing the representation learnt by a neuron instead of the manifold formed by the activation vectors of the same type. Instead, what's been proposed in our work is actually the comparison of the content learnt by two deep neural networks. We are aiming at finding the features and knowledge difference between two networks. The algorithm and criteria in our work can be generalized to the comparison between any two neural networks. Especially when two neural networks have the same input and similar accuracy, the curvature-based criteria can help in telling the detail of possible different learning methods of these networks.

Finally, as for data augmentation methods, we adopt SVD-based method to generate close images. The work in [6], [7] and also in [1], mainly follow the idea of injecting noise into the input of a neural network during image generating. The intension of these methods is to enlarge the dataset using label-preserving transformations and thus does not take the closeness of activation vectors into account. The goal of the images generating in our work, however, is to create a large amount of data which is not only lable-preserving but also close to the original activation vector. Moreover, our methods can be roughly seen as removing noise and the efficiency has been shown by the experiment.

V. Conclusion

In this paper, two neural networks with the same structure but different parameters are compared. We aim at finding the shared knowledge, different learning methods between two networks with the help of intrinsic geometry of the manifolds embedded in them. To fulfill this goal, we have developed an effective close-data generating strategy. Then the dimensions of manifolds are calculated and it turns out that they are the same for two AlexNets.

As traditional Euclidean-distance based method cannot explain the existence of both similarity and difference between two AlexNets, our major contribution is that we take Riemann curvature distribution and sectional curvature distribution to be the criteria for the comparison of two neural networks. Our experiments show that though the two AlexNets have similar accuracy rate, their manifolds behave differently for certain inputs. We believe this brings more insight into the intrinsic learning behaviors of deep neural networks. It should be mentioned that the work in this paper focus on building similarity and difference between two networks, to figure out the specific meaning of these similarities, differences and global properties will be our next work.

The reader may have noticed that in this paper we only work on the manifolds for the last three fully connected layers though AlexNet has convolutional layers as well. One way is to repeat the above work for convolutional layers by taking every output of them as a long vector as in [4]. However to our opinion, unlike the fully connected layers which is straightforward to be treated as vectors in high dimension, convolutional layer is more of matrix type, where the vector-form feature descriptor may break the geometric structure of the pixel matrix space. Redefine manifold for convolutional layers is also part of our future work.

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